



Supermarket Customer Behavior Analysis



Understanding Spending Patterns and Campaign
Performance

Analysis using Supermarket Customer Dataset
Capstone Project Module 3
Haplila - May 2026



Executive Summary

Key Customer Insights & Business Findings



Key Metrics Overview

Total Customers

2,237

Total Spending

1.4M

Campaign Response Rate

15%

Complaint Rate

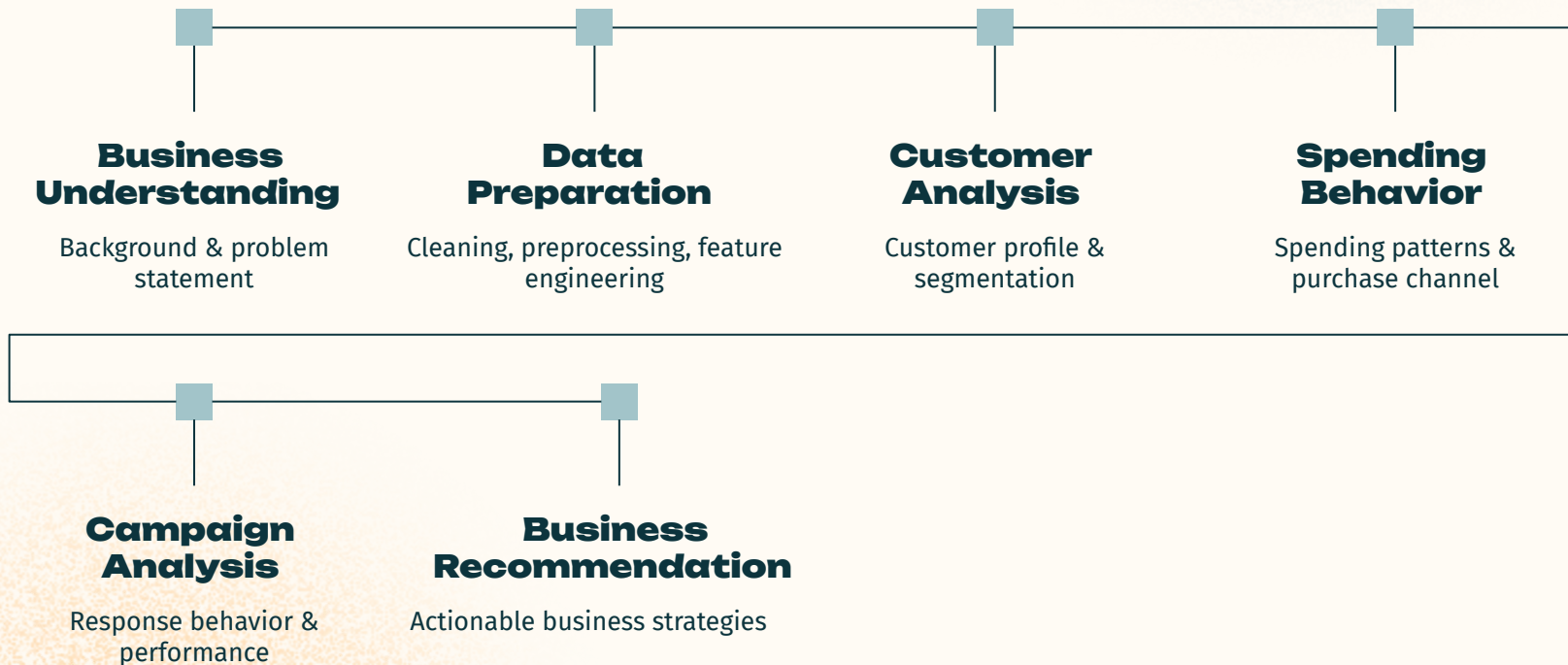
1%

- Medium value customers account for ~50% of total customers.
- Wine products contribute the highest spending (~680K).
- Store is the dominant purchase channel.
- High-value customers show stronger campaign response.
- Customer purchases are not heavily driven by discounts.



PRESENTATION ROADMAP

Supermarket Customer Analysis Flow



Business Needs Better Understanding of Customer Behavior

Understanding customer characteristics and campaign performance can improve business decisions

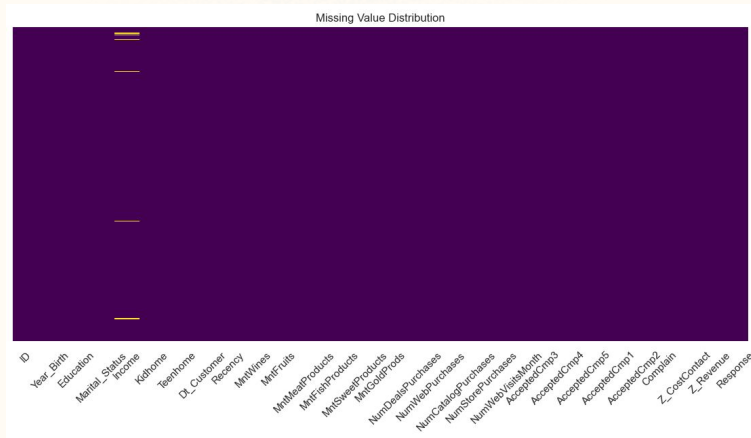
Latar Belakang

- Supermarket stores customer demographic and purchase data
- Data includes income, products purchased, campaign response, and complaints
- Understanding customer behavior helps identify:
 - valuable customers
 - spending patterns
 - effective campaign targets

Business Questions

1. Who are the supermarket customers?
→ Understand customer characteristics
2. Which customer segments contribute the highest value?
→ Identify valuable customer groups
3. Which products generate the highest spending?
→ Identify key revenue contributors
4. Which customers respond better to campaigns?
→ Improve campaign effectiveness

Dataset Required Data Cleaning



Missing Value Insight

- Missing values found in Income
- Income may contain outliers
- Median imputation selected because it is more robust than mean

Missing Value Handling:

```
income_median = df['Income'].median()
df['Income'] = df['Income'].fillna(income_median)
```

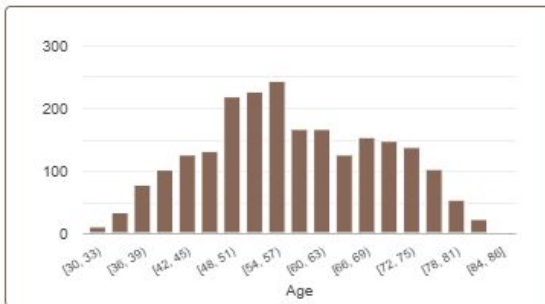


Cleaning Result

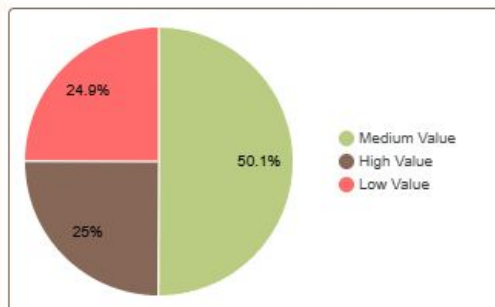
- ✓ Imputation value: 51,381.5
- ✓ Missing values after treatment: 0

Medium-Value and Middle-Aged Customers Dominate the Customer Base

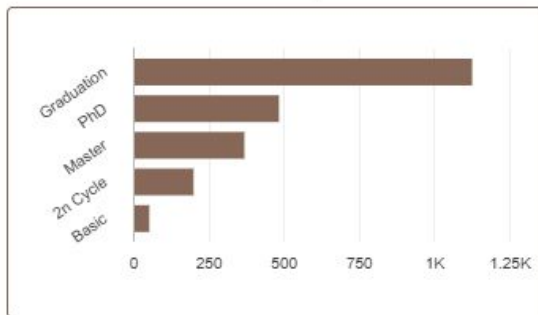
Age Distribution



Customer Value Segment



Customer Distribution by Education

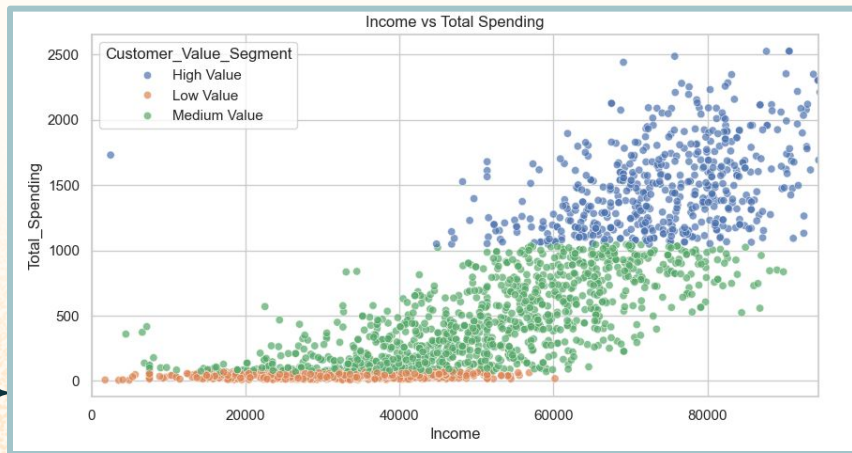


Customer Insights:

- ✓ Medium-value customers account for ~50%
- ✓ Majority customers are middle-aged
- ✓ Graduation is dominant

Wine Products Generate the Largest Customer Spending

Customer spending increases with income, although several income outliers exist



Spending by Product Category

Wines
680.0K

Meat Products
373.4K

Gold Products
98.4K

Fish Products
83.9K

Sweet Products
60.6K

Fruits
58.8K

Key Insights:

- ✓ Wine products contribute the highest spending (~680K)
- ✓ Higher-income customers generally exhibit higher spending behavior
- ✓ ✓ Several high-income outliers may influence spending interpretation

Store Purchases Dominate While Discount Purchases Remain Low

Customers prefer direct purchasing channels and show limited dependence on discounts



Discount Purchase Analysis



Purchases by Channel

WEB
9.1K

CATALOG
6.0K

STORE
13.0K

Key Insights:

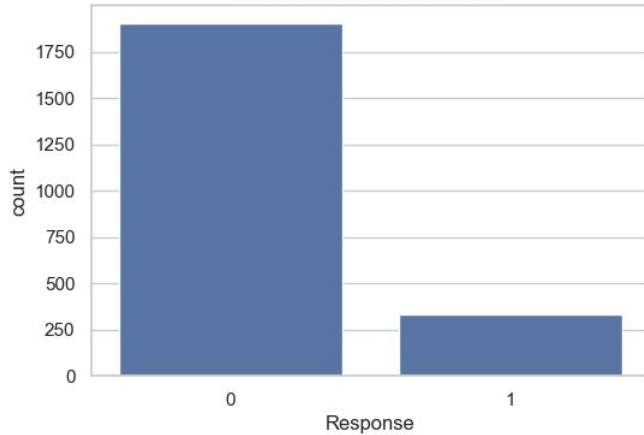
- ✓ Store purchases contribute the highest transaction volume (~13K)
- ✓ Web and catalog purchases are lower than store purchases
- ✓ Most customers rarely purchase using discounts
- ✓ Purchasing behavior appears driven more by preference than promotions

High-Value Customers Show Stronger Campaign Response

Customer response differs across value and activity segments

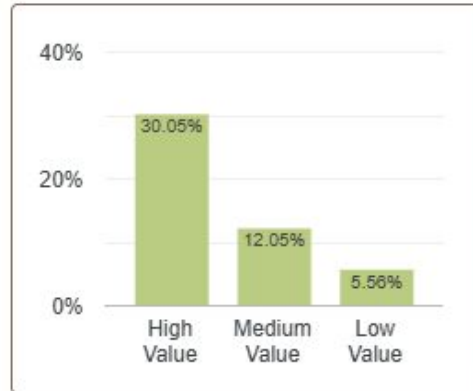


Distribusi Campaign Response

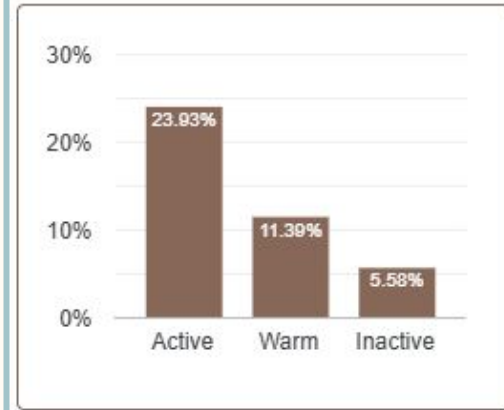


Response = 0 → No Response
Response = 1 → Responded

Response by Customer Segment



Response by Recency Segment

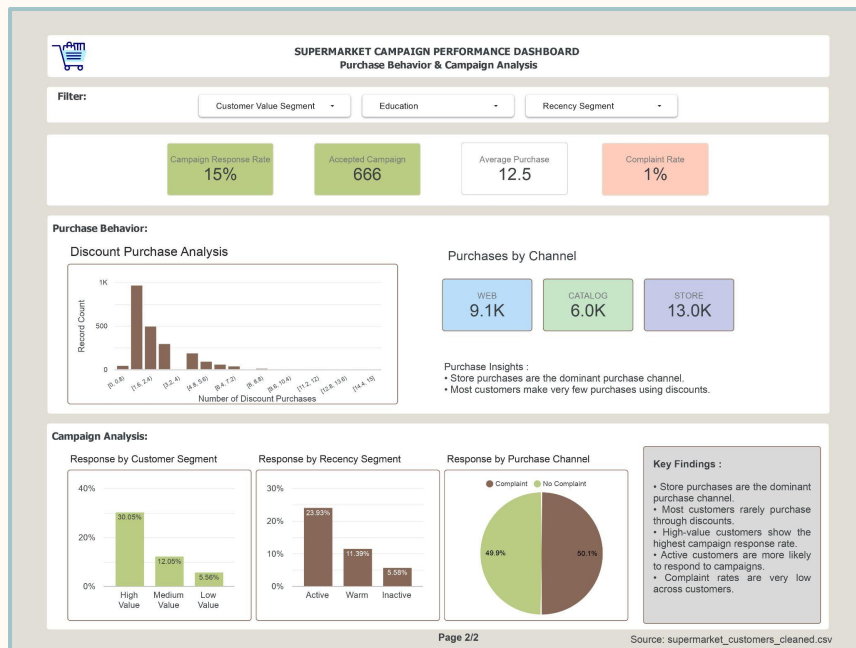


Key Insights:

- ✓ Most customers do not respond to campaigns
- ✓ High-value customers show the strongest response
- ✓ Active customers are more likely to respond
- ✓ Campaign targeting can be improved using segmentation

Interactive Dashboard Enables Faster Customer Exploration and Decision-Making

User can explore data dynamically using interactive filters and visual summaries



Customer Insights Can Be Translated into Actionable Business Strategies

Recommendations are developed based on customer behavior and campaign findings

FINDINGS	ACTIONS	EXPECTED IMPACT
High-value customers show stronger campaign response	Create personalized campaigns and loyalty offers	Increase campaign effectiveness and conversion rate
Store purchases dominate (~13K)	Improve loyalty program & in-store experience	Higher retention
Customer rarely use discounts	Apply targeted promotions	Better promotion efficiency
Active customers respond better	Create re-engagement campaigns for inactive customers	Improve customer retention

Customer Insights Can Improve Marketing Effectiveness and Business Decisions

Customer Insights



Medium-Value customers dominate.

Spending Insights



Wines drives highest spending.

Campaign Insights



High-value customer respond.

Final Takeaway:

Customer segmentation and spending analysis can support more targeted marketing and improve campaign efficiency.